Internship program - 2026

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Proposed internship timeline

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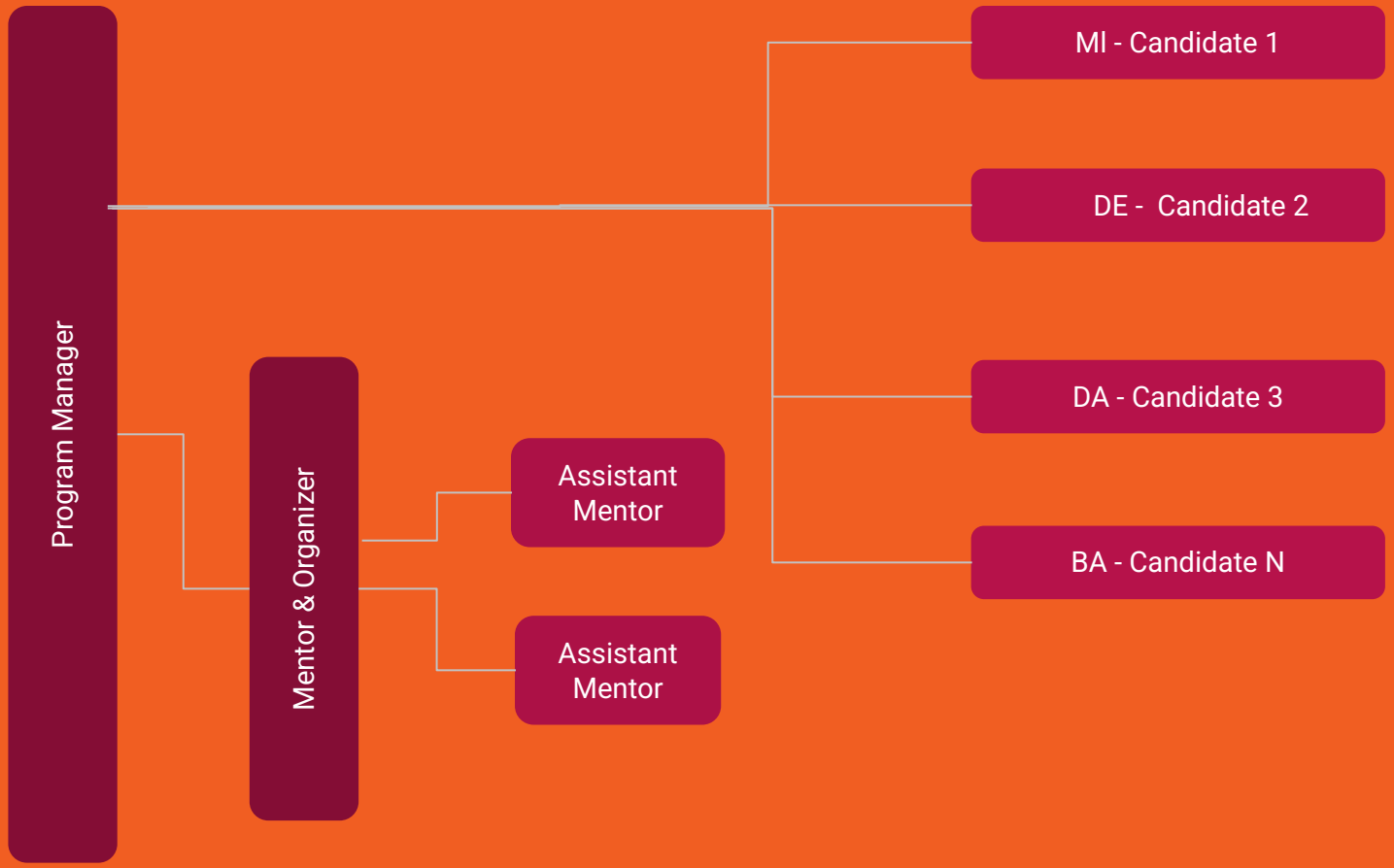
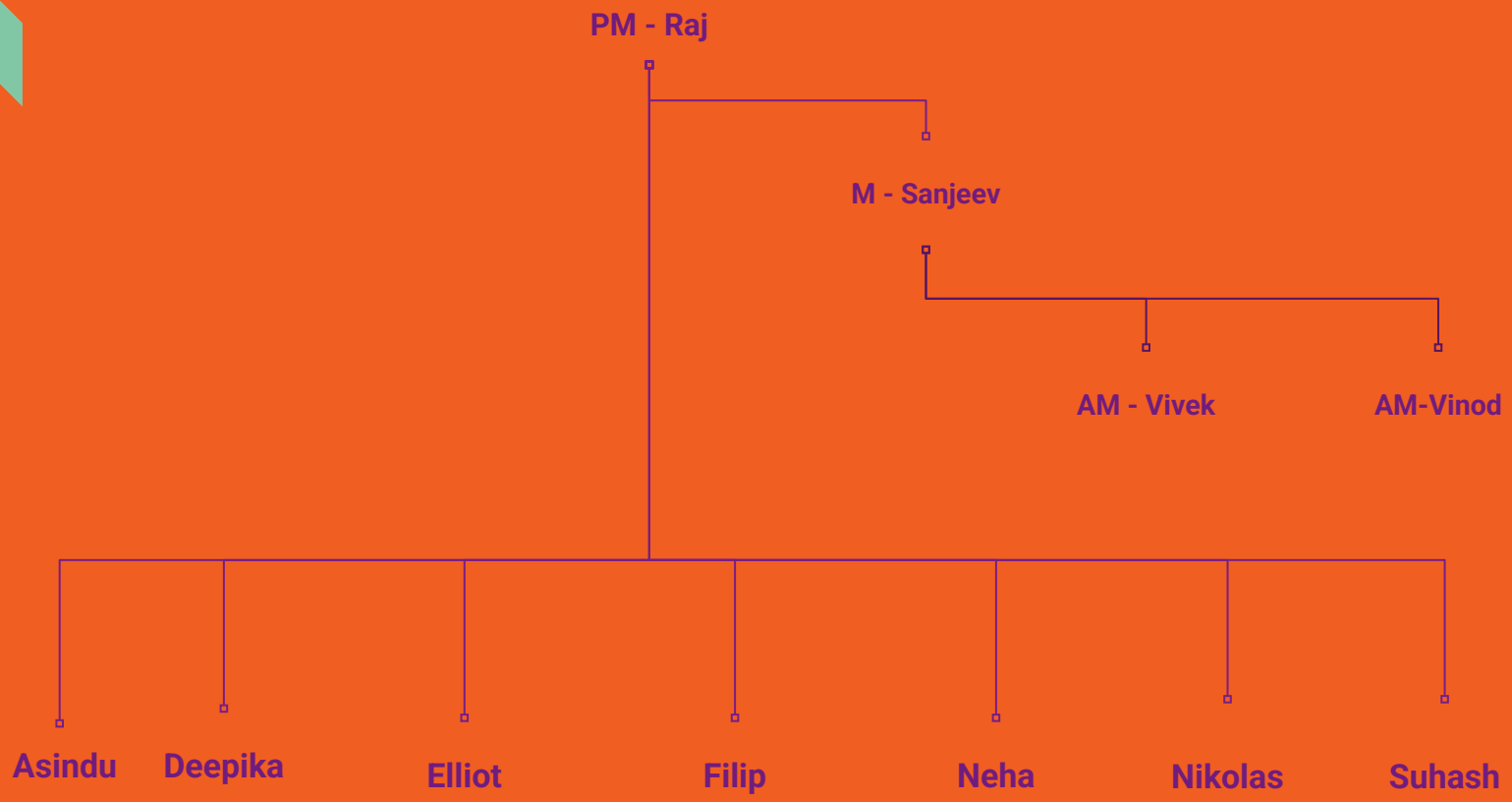
Program Reporting hierarchy



Team hierarchy



Business Intelligence / Data Analyst

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Month 1: The Core Querying & Logic

- **SQL (Weeks 1-2):** This is your most used tool. Master JOINs, GROUP BY, and **Window Functions**. Practice daily on HackerRank or SQLZoo.
    - SELECT, WHERE, GROUP BY
    - JOIN (deep understanding)
    - Subqueries
    - CTE
    - Window functions
    - Indexes
    - Query optimization basics
- **Basic Statistics (Week 3-4):** Learn descriptive statistics (mean, variance, correlation) and inferential basics like **P-values** and **Hypothesis Testing**.
    - Mean, median, mode
    - Variance & standard deviation
    - Probability basics
    - Correlation
    - Normal distribution
    - Hypothesis testing (basic idea)
    - Window functions (ROW_NUMBER, RANK)
    - CASE statements
    - CTEs
    - Aggregation problems
    - Real business queries

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Business Intelligence / Data Analyst - Page 2

Month 2: Data Manipulation with Python

- **Python Basics (Weeks 5-6):** Focus on data-specific syntax. Learn lists, dictionaries, and loops.
- **The Power Trio (Weeks 7-8):** Master Pandas for data cleaning, NumPy for math, and Matplotlib/Seaborn for basic charting.
- **Excel (Weekend Deep-Dive):** Ensure you are an expert in **Pivot Tables** and XLOOKUP.

Month 3: Visualization & Business Intelligence (BI)

- **BI Tools (Weeks 9-11):** Pick **one**—either Tableau or Microsoft Power BI. Learn to build interactive dashboards that highlight **KPIs** (Key Performance Indicators).
- **Data Storytelling (Week 12):** Learn how to present findings. A good analyst doesn't just show a graph; they explain the *business impact* and suggest an action.

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Business Intelligence / Data Analyst - Page 3

Month 4 to 5 : Portfolio & Professional Presence

- **The Capstone (Weeks 13-14):** Find a "messy" dataset on Kaggle. Clean it in Python, query it in SQL, and build a dashboard in Power BI/Tableau.
- **GitHub (Week 15):** Document your project. Use GitHub for your code.
- **Certification**

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Machine Learning Engineer

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ML Engineer - Page 1

Month 1: Python, Math & ML Fundamentals

- **Advanced Python:**
  - Master decorators, generators, and OOP.
  - Learn Fast API to wrap models into web services (APIs).
- **Core ML Libraries:** Become proficient in Scikit-learn for classical ML and Pandas for data engineering.
- **Linear Algebra & Stats:** Refresh your knowledge of matrix multiplication and probability—essential for understanding how models actually "learn."

Month 2: Deep Learning & Frameworks

- **Neural Networks:** Learn the architecture of CNNs, RNNs, and Transformers.
- **Frameworks:** Choose either PyTorch or TensorFlow. Focus on building, training, and saving model weights.
- **Version Control for Data:** Learn DVC (Data Version Control) to track changes in datasets just like you track code with Git.

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ML Engineer - Page 2

Month 3: Cloud Platforms & Model Deployment

Pick a platform: **AWS** (SageMaker), **GCP** (Vertex AI), or **Azure** (Azure ML) or **Databricks** (ML Flow).

- **Containerization:** Master **Docker**. You must be able to "containerize" your model so it runs anywhere.
- **Model Serving:** Learn how to deploy models using Triton Inference Server or simple Docker containers on cloud instances.
- **Inference:** Understand the difference between **Batch Inference** (running predictions on a schedule) and **Real-time Inference** (instant results via API).

Month 4 - 5 : Implement a project including MLOps & Production Scaling

- **ML Pipelines:** Learn to automate the training-to-deployment flow using Kubeflow or MLflow.
- **Monitoring:** Learn to track **Model Drift** (when a model's accuracy drops over time in production).
- **Capstone Project:**
  1. Train a model (e.g., Image Classification or Sentiment Analysis).
  2. Containerize it with **Docker**.
  3. Deploy it to a **Cloud Provider** (AWS/GCP/Azure) using an automated pipeline.
- **Certification:** Target the Google Professional Machine Learning Engineer or AWS Certified Machine Learning – Specialty.

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Cloud/DevOps Engineer

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Cloud/DevOps Engineer - Page 1

Month 1: Linux, Networking & Scripting

The cloud is just "someone else's computer," and that computer almost certainly runs Linux.

- Linux (Weeks 1-2):
  - Master the CLI.
  - Learn permissions,
  - process management, and the boot process. Follow the Linux Journey curriculum.
- Networking (Week 3):
  - Understand VPCs,
  - Subnetting, Route Tables,
  - DNS, and
  - the difference between TCP and UDP.
- Python/Bash (Week 4): Learn to automate boring tasks. Write a script that backups a directory or checks if a website is down.

Month 2: The Cloud Platform & Containers

Pick AWS (widely used) or Azure (enterprise favorite).

- Cloud Core (Weeks 5-6):
  - Learn Identity Access Management (IAM),
  - Compute (EC2/VMs), and
  - Storage (S3/Blob).

  Practice via the Free Tier.

- Docker (Weeks 7-8):
  - Learn to containerize applications.
  - Understand images, layers, and volumes.
  - Use Docker Desktop for local practice.

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Cloud/DevOps Engineer - Page 2

Month 3: Infrastructure as Code (IaC) & CI/CD

This is where you become a "DevOps" engineer rather than just a SysAdmin.

- Terraform (Weeks 9-10):
  - Learn to define your entire cloud setup as code.
  - Use HashiCorp Learn for guided labs.
- CI/CD Pipelines (Weeks 11-12):
  - Build automation with GitHub Actions or GitLab CI.
  - Create a pipeline that automatically tests and deploys code when you push to GitHub.
- Kubernetes (Weeks 12):
  - Learn the basics of orchestration—pods, deployments, and services.
  - Don't go too deep; just learn how to deploy an app.
  -

Month 4: Implement a project including Security & Portfolio also

- Security & Monitoring:
  1. Learn to use tools like Prometheus for monitoring and Grafana for visualization.
- The "Resume Maker" Project:
  1. Provision an AWS/Azure environment using Terraform.
  2. Deploy a containerized app using Docker.
  3. Automate the whole thing with a CI/CD pipeline.
- Certification: Finalize your studies for the AWS Certified Solutions Architect – Associate or AZ-104.

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Data Engineer - Page 1

Month 1: The Core Duo (SQL & Python)

- **SQL Mastery:** Move beyond basic queries. Master **Window Functions**, Common Table Expressions (CTEs), and complex Joins. Practice on SQL Murder Mystery or HackerRank.
- **Python for Data:** Focus on data structures (dictionaries, lists) and libraries like **Pandas** for manipulation and **Requests** for API integration.
- **Development Workflow:** Learn **Git** for version control and basic **Linux/Bash** commands for navigating servers.

Month 2: Data Architecture & Modeling

- **Databases:** Understand the difference between **OLTP** (PostgreSQL) and **OLAP** (Snowflake/BigQuery).
- **Modeling:** Learn **Star Schema** and **Snowflake Schema** design. This is critical for making analytical queries run 10x faster.
- **Cloud Foundation:** Pick **one** major cloud provider (**AWS, Azure, or GCP**) and learn its core storage (e.g., AWS S3) and compute (e.g., Lambda) services.
- **Emerging AI Skills:** Learn to build **RAG pipelines** and work with **Vector Databases** (like Pinecone or pgvector) to support AI applications.

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Month 3: Pipelines & Orchestration

- **ETL/ELT Frameworks:** Use **dbt (data build tool)** to transform data using SQL and manage it like code.
- **Orchestration:** Learn Apache Airflow to schedule, monitor, and automate your data workflows.
- **Distributed Processing:** Get hands-on with **PySpark** (the Python API for Apache Spark) to handle datasets larger than a single machine's memory.

Month 4 - 5 : AI Integration & Portfolio

- **Capstone Project:** Build an end-to-end pipeline:
  1. Scrape data or call an API using **Python**.
  2. Ingest it into a **Cloud Data Warehouse**.
  3. Transform it using **dbt**.
  4. Orchestrate the whole process with **Airflow**
  5. **Engineering Certification**

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All the best!

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